Combined, paraphrased and repeated GI descriptions

Exploratory follow-up | 30 October 2026

Design

We tested concatenation of all four earlier descriptions, that concatenation twice, compact text twice, and three new paraphrases each once or exactly twice. Biological facts were held constant: human peripheral-blood cell identity, unstimulated state and 10x 3-prime single-cell RNA-seq. Model, DNA, TSS, allele orientation and 40,960-bp effective geometry were unchanged. REF and ALT used identical text.

The same 16 development pairs were used for selection. Correct directions divided by all 16 pairs was the objective; zero effects count as incorrect. The incumbent compact description is retained on primary ties. Spearman correlation breaks ties only among strictly better alternatives. A new 20-locus validation set excludes all previously tested loci, including the first optimization holdout, using nonoverlapping ±100-kb TSS windows. No fresh predictions were evaluated before selection froze.

Development results

Description	Correct / 16	Nonzero	Spearman rho
baseline	3	13	-0.044
compact	7	13	0.516
combined_once	7	13	0.404
combined_twice	6	13	0.369
compact_twice	7	14	0.422
paraphrase_assay_first_once	5	12	0.236
paraphrase_assay_first_twice	6	13	0.304
paraphrase_cell_first_once	6	12	0.237
paraphrase_cell_first_twice	6	13	0.106
paraphrase_sentence_once	6	12	0.321
paraphrase_sentence_twice	9	14	0.444

Frozen winner: **paraphrase_sentence_twice**. Searching nine alternatives on sixteen pairs is highly susceptible to development-set overfitting; the complete table includes unsuccessful descriptions.

Does exact repetition help?

combined: correct directions 7 → 6 /16; Spearman rho 0.404 → 0.369.
compact: correct directions 7 → 7 /16; Spearman rho 0.516 → 0.422.
paraphrase_assay_first: correct directions 5 → 6 /16; Spearman rho 0.236 → 0.304.
paraphrase_cell_first: correct directions 6 → 6 /16; Spearman rho 0.237 → 0.106.
paraphrase_sentence: correct directions 6 → 9 /16; Spearman rho 0.321 → 0.444.

Direct once-versus-twice checks on new loci

combined: 11/20 correct once versus 9/20 twice; paired all-pair change -10.0 pp (95% CI -30.0 to 10.0). These diagnostics do not re-select the frozen prompt.
paraphrase_sentence: 9/20 correct once versus 10/20 twice; paired all-pair change 5.0 pp (95% CI -10.0 to 20.1). These diagnostics do not re-select the frozen prompt.

Frozen-choice validation and original cohort

At the 20 new loci, baseline versus selected correct directions were 10/20 versus 10/20. The all-pair gain was 0.0 percentage points (95% paired locus-bootstrap interval -15.0 to 15.0). Conditional direction agreement was 83.3% versus 66.7%; abstentions were 8 versus 5. These small samples do not establish general improvement when the paired interval includes zero.
On the already-reported 100-pair cohort, pooled Spearman rho changed from 0.370 to 0.442; conditional direction agreement from 67.5% to 75.0%. All-pair correct directions changed from 52/100 to 60/100. This is post-tuning description, not an untouched validation set.

Next strategies (proposed, not executed)

Use a controlled ablation matrix: gene symbol; Ensembl gene ID; both; symbol-only repetition; canonical cell name; verified Cell Ontology ID; and explicit assay fields. Add one field at a time and keep REF/ALT text identical. Gene identity may enable learned gene lookup and must be analyzed separately from sequence-only conditioning. Freeze choices on development data before using another new validation cohort.

Selected description

Homo sapiens peripheral-blood {cell}, unstimulated. Single-cell RNA sequencing with 10x Genomics Chromium 3-prime poly(A) capture; source material: peripheral blood mononuclear cells (PBMCs). Homo sapiens peripheral-blood {cell}, unstimulated. Single-cell RNA sequencing with 10x Genomics Chromium 3-prime poly(A) capture; source material: peripheral blood mononuclear cells (PBMCs).

Interpretation and audit

Repetition is a testable conditioning change, not a guaranteed way to improve prediction. A development gain should be distinguished from fresh-locus gain, and higher correlation from better directional accuracy. Multiple prompt searches, small validation sets, rounding/abstentions, observational eQTLs and unknown training overlap limit interpretation. The original zero-shot benchmark remains unchanged. Exact strings for all descriptions and measured results accompany this PDF.
API preflight: the user identified themselves as the API developer and confirmed this key’s quota was lifted. The software retained a bounded cumulative ceiling of 1,600 attempts and serial requests; headers were logged, with no automatic retries. This round made 656 new attempts; cumulative attempts including the initial preledger call were 1444. Monetary pricing was not published and no dollar amount is claimed. Reproduce: python scripts/optimize_context_round2.py; python scripts/build_repetition_pdf.py (GI_API_KEY in environment). Data: doi:10.5281/zenodo.18221260.